

ShopCo Profit Growth Dashboard

An interactive version of the profit-growth case for five teams. One shared overview, then one page per problem area. Pages are organized by problem, not by team, so teams that share a problem look at the same numbers.

“ShopCo has been flat for five years because it stopped acquiring customers, and it gives away \$2.8M a year in discounts that don't change what customers buy.”

Who it serves

TEAM	NEEDS TO KNOW	PAGES
Finance	Are we on track for +20%? Where is margin leaking? What is each lever worth?	0 · 1 · 3 · 5
Sales	What drives revenue? Do promotions lift volume?	0 · 1 · 4
Growth	Are we adding and keeping customers? Value vs. cost?	0 · 2 · 5
Product	Where do purchases fail? Where is the experience bad?	3
Marketing	Which channels acquire customers? Do loyalty discounts work?	1 · 2 · 4

Pages

PAGE	TITLE (THE FINDING)
Page 0 · Executive Overview	Flat for five years: the customer pipeline has dried up
Page 1 · Pricing & Discounts	Discounts under 23% cost \$2.8M a year and don't change buying
Page 2 · Customer Growth	New customers fell 96% since 2022: we're living off the 2021 base
Page 3 · Revenue Leakage	Every cancelled order is a failed payment: \$1.0M of profit lost a year
Page 4 · Channels & Markets	Channel and market mix isn't the problem: profit per order varies less than 3%
Page 5 · Scenario Planner	Four levers can deliver +\$3.1M a year against a \$2.6M target
Page 6 · About the Data	Definitions, sources, refresh date, caveats

Data model and standards

Data: six prepared CSVs (`prepare_tableau_data.py`)

Relationships: **order_lines** —order_id— **orders** —customer_id— **customers** —customer_id— **customer_year**; **products** joins **order_lines** on product_id.

Revenue, merch profit and discount band are precomputed. Build ratios (margin, rates) as calculated fields so they stay correct under filters.

FILE	GRAIN	ROWS	KEY FIELDS
<code>order_lines</code>	Order line	397,569	<code>discount_pct</code> , <code>discount_band</code> , <code>quantity</code> , <code>revenue</code> , <code>merch_profit</code> , <code>in_discount_analysis</code>
<code>products</code>	Product	1,175	Name, supplier, list price, rating
<code>orders</code>	Order	138,116	Date, status, channels, payment, delivery, rating, order totals, <code>is_first_order</code> , <code>is_payment_failure</code> , <code>is_returned</code> , <code>is_late</code>
<code>customers</code>	Customer	25,000	Segment, country, <code>customer_acquisition_cost</code> , <code>cohort_year</code> , lifetime totals
<code>customer_year</code>	Customer × year	83,730	<code>cohort_year</code> , <code>is_new</code> , <code>active_prior_year</code> , yearly revenue and profit
<code>lever_summary</code>	Lever	4	Separate data source for a static waterfall

Layout

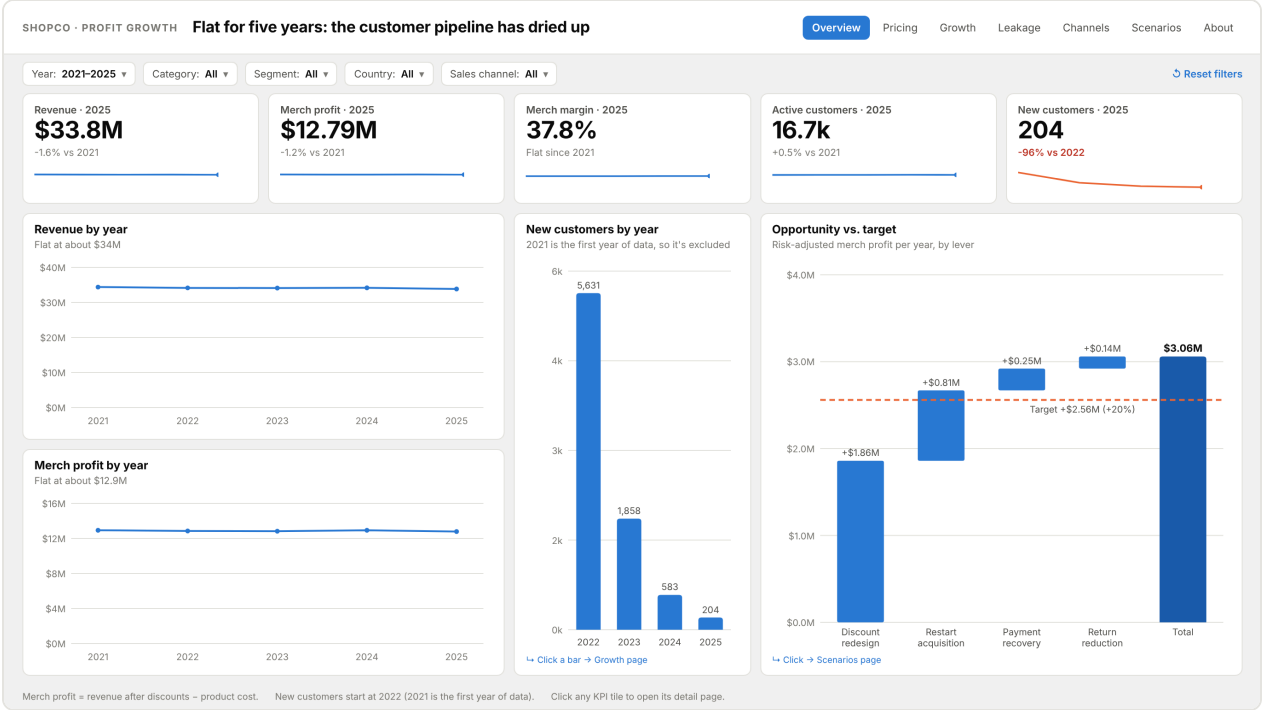
- Fixed **1366 × 768**. Header with the finding as the title, plus navigation; one filter row; a footnote line.
- Global filters: **Year, Category, Segment, Country, Sales channel**, applied to all sheets using the data source (not Scenarios).
- A reset button on every page. Drill Year → Quarter → Month on time axes.

Visual rules

-  **Primary**: default bars and lines `#2a78d6`
-  **Comparison**: second measure, target lines `#eb6834`
-  **Context**: non-selected, 'other' `#b4b2ab`
- **Sequential**: heatmaps and map, light → dark `#cde2fb` → `#0d366b`
- **Status**: good / bad KPI states only, always with ✓ / ▲ icon `#008300` / `#c98500`
- No dual-axis charts. Legend whenever there are 2+ series. Light gridlines. `$#,##0.0,, "M"` for millions.

Flat for five years: the customer pipeline has dried up

One screen everyone starts from. It shows that the top line is flat and the reason is acquisition, then points to the fix.



SHEET	CHART	FIELDS
KPI tiles ×5	Big number + sparkline	order_revenue , order_merch_profit , margin, COUNTD(customer_id) , new = is_first_order ; 2025 vs. first year
Revenue / Merch profit by year	Two line charts (no dual axis)	order_year × SUM(order_revenue) / SUM(order_merch_profit) , y-axis from 0
New customers by year	Bars	orders where is_first_order , order_year ≥ 2022
Opportunity vs. target	Waterfall (Gantt bars)	lever_summary.risk_adjusted ; target line = 20% × 2025 merch profit

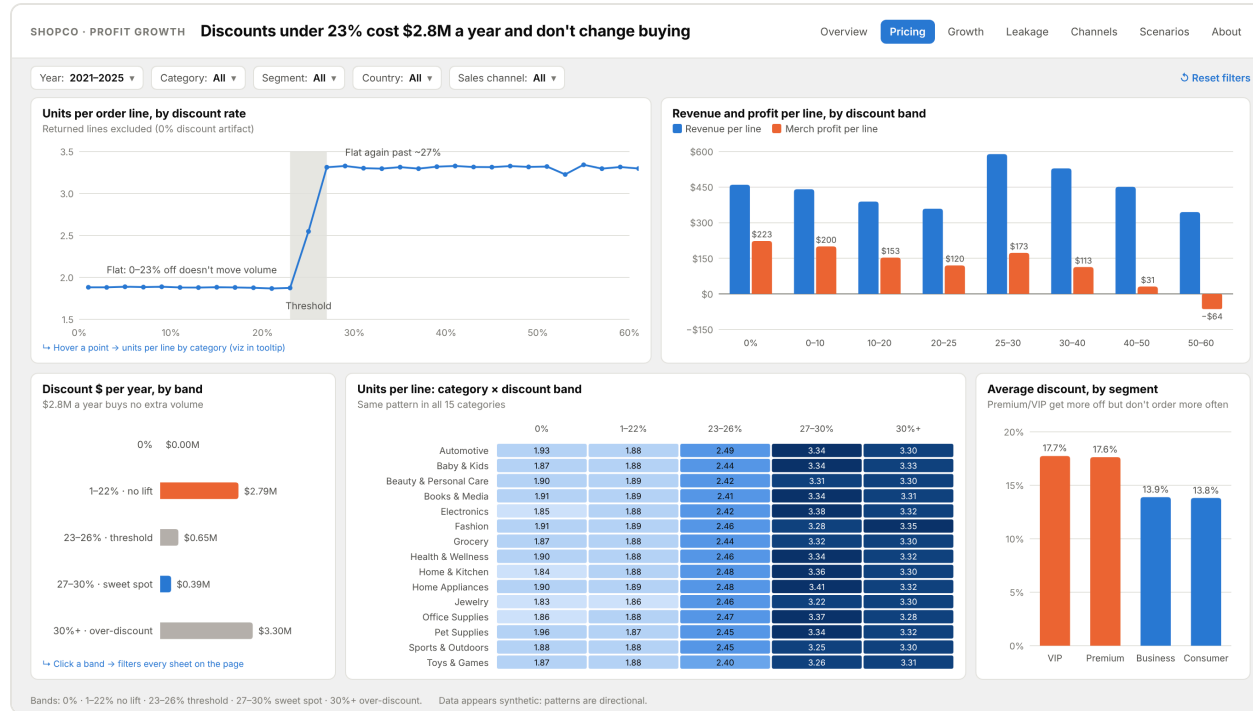
- INTERACTIONS
- KPI tile → go to its detail page
 - New-customer bar → Growth page
 - Waterfall → Scenarios page

VALIDATE (ALL FILTERS CLEARED)

2025: \$33.8M revenue · \$12.79M merch profit · 37.8% margin · 204 new customers

Discounts under 23% cost \$2.8M a year and don't change buying

Shows the demand threshold, what each discount level earns, and how much discount goes where it buys nothing.



SHEET

Units per line by discount rate

CHART

Line + reference band 23-27%

FIELDS

`discount_pct bin (0.02) × AVG(quantity) ; filter in_discount_analysis = True`

Revenue & profit per line by band

Side-by-side bars, one \$ axis

Custom group on `discount_pct` (0, 0-10, 10-20, 20-25, 25-30, 30-40, 40-50, 50-60) × `AVG(revenue)`, `AVG(merch_profit)`

Discount \$ per year by band

Horizontal bars

`discount_band × SUM(discount_amount) / 5`; highlight the no-lift band

Category × band

Highlight table, sequential blue

`product_category × discount_band`, `AVG(quantity)`

Discount by segment

Bars

`customer_segment × SUM(order_discount)/SUM(order_gross_sales)`

INTERACTIONS

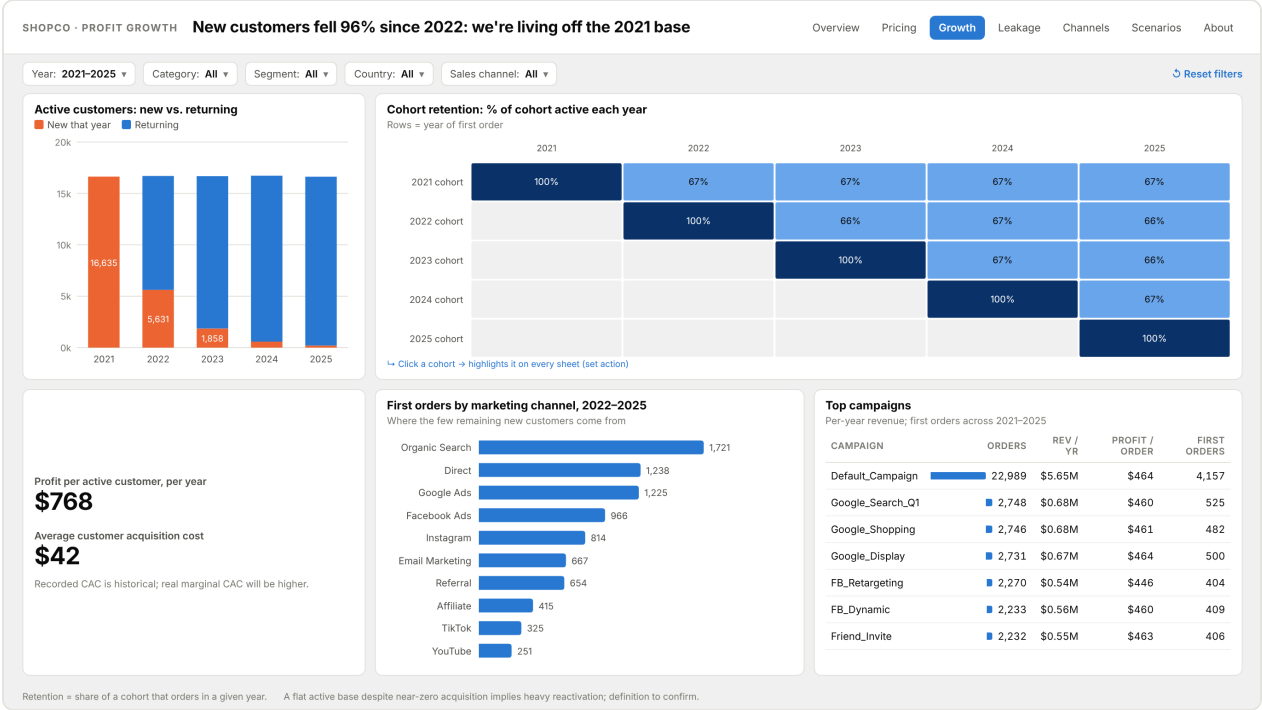
- Click a band → filters every sheet on the page
- Hover the curve → viz in tooltip: units per line by category

VALIDATE (ALL FILTERS CLEARED)

Units per line 1.88 (0-20%) / 3.31 (30%+) · no-lift discount \$2.79M per year

New customers fell 96% since 2022: we're living off the 2021 base

Shows the acquisition collapse, retention by cohort, and what a customer is worth vs. what they cost.



SHEET	CHART	FIELDS
New vs. returning active customers	Stacked bars	<code>customer_year : year × COUNTD(customer_id) , color by is_new</code>
Cohort retention	Highlight table, % of cohort	<code>cohort_year × year ; COUNTD(customer_id) / cohort size (FIXED on cohort_year)</code>
Value vs. cost	Two KPI numbers	AVG of yearly merch_profit per customer; AVG(customer_acquisition_cost)
First orders by marketing channel	Sorted horizontal bars	<code>orders where is_first_order and order_year ≥ 2022 × marketing_channel</code>
Top campaigns	Text table with in-cell bars	<code>campaign_name : orders, revenue / 5, profit per order, first orders</code>

INTERACTIONS

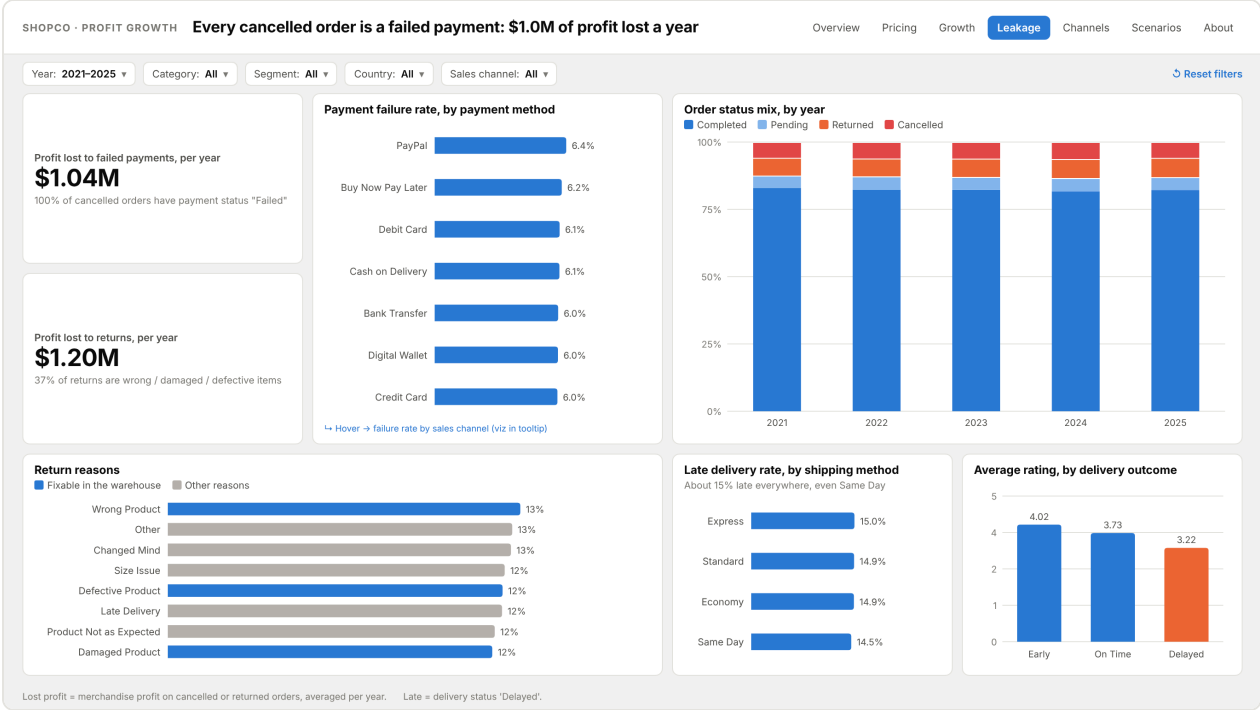
- Click a cohort → set action highlights it on every sheet
- Year filter drives every sheet except the cohort table

VALIDATE (ALL FILTERS CLEARED)

New customers 5,631 → 1,858 → 583 → 204 · retention ≈ 67% · \$768 profit per active customer · \$42 CAC

Every cancelled order is a failed payment: \$1.0M of profit lost a year

Sizes the profit lost to failed payments and returns, and flags the late-delivery experience problem.



SHEET	CHART	FIELDS
Lost-profit tiles	KPI numbers	SUM(order_merch_profit) / 5 where is_payment_failure ; same for is_returned
Payment failure rate	Horizontal bars	payment_method × AVG(INT(is_payment_failure))
Order status mix	100% stacked bars	order_year × order_status , % of orders
Return reasons	Sorted bars, 2 colors	return_reason ; blue = wrong / damaged / defective, gray = other
Late delivery & rating	Bars + bars	shipping_method × SUM(INT(is_late))/SUM(INT(is_delivered)) ; delivery_status × AVG(customer_rating)

INTERACTIONS

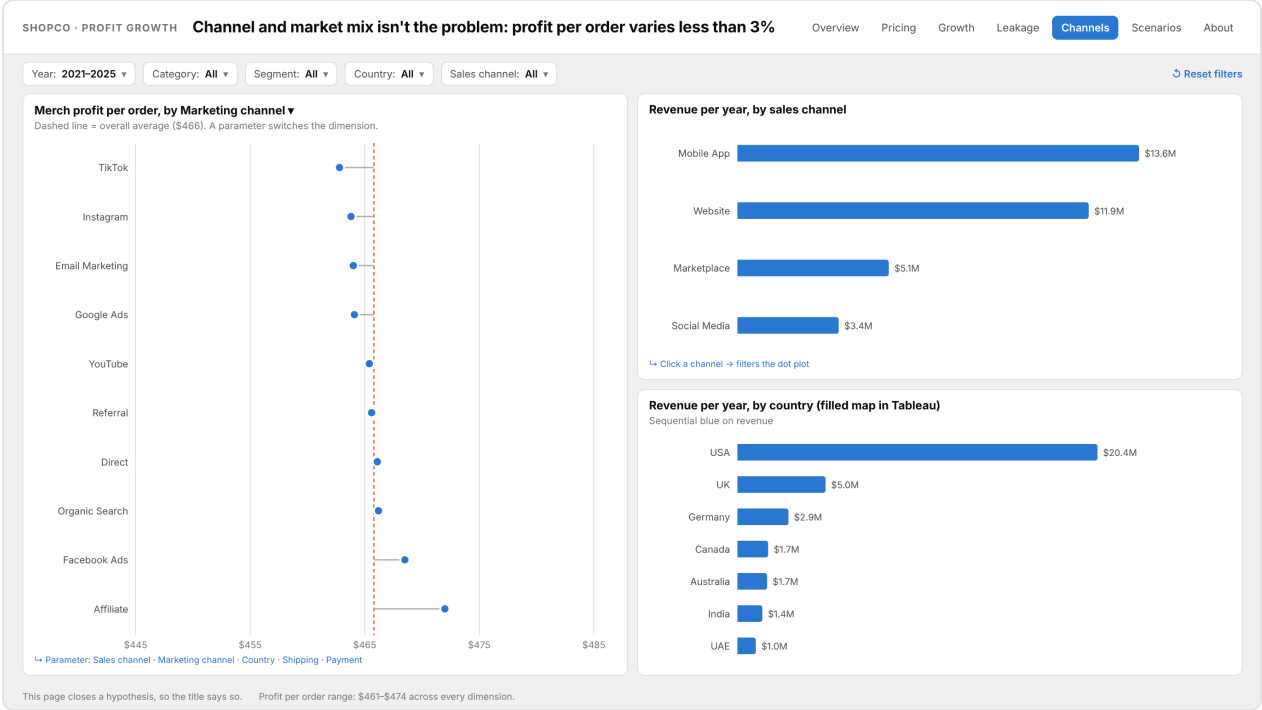
- Hover payment bars → viz in tooltip: failure rate by sales channel
- Click a year in the status chart → filters the page

VALIDATE (ALL FILTERS CLEARED)

Lost profit per year: \$1.04M cancelled · \$1.20M returned · 15% late deliveries

Channel and market mix isn't the problem: profit per order varies less than 3%

Closes a hypothesis so teams stop spending time on it. The title says so on purpose.



SHEET	CHART	FIELDS
Profit per order by dimension	Dot plot + average reference line	Parameter <code>p_Dimension</code> (Sales channel / Marketing channel / Country / Shipping / Payment) → CASE field × <code>AVG(order_merch_profit)</code>
Revenue by sales channel	Horizontal bars	<code>sales_channel</code> × <code>SUM(order_revenue) / 5</code>
Revenue by country	Filled map, sequential blue	<code>customer_country</code> × <code>SUM(order_revenue) / 5</code>

INTERACTIONS

- Parameter switches the dot-plot dimension
- Click a sales channel → filters the dot plot

VALIDATE (ALL FILTERS CLEARED)

Profit per order \$461–\$474 across every dimension

SHOPCO - PROFIT GROWTH

Four levers can deliver +\$3.1M a year against a \$2.6M target

Overview
Pricing
Growth
Leakage
Channels
Scenarios

Global filters don't apply on this page

Assumptions

Tableau parameters

Discount floor 23%

Discount cap 30%

Payment recovery 30%

Return reduction 20%

New customers per year 1,000

Confidence by lever

Pricing 50%

Acquisition 50%

Payments / Returns 80% / 60%

Risk-adjusted merch profit per year, by lever

Updates live as the parameters change

Lever	Value
Discount redesign	+\$1.86M
Restart acquisition	+\$0.81M
Payment recovery	+\$0.25M
Return reduction	+\$0.14M
Total	\$3.06M

Risk-adjusted total vs. target

\$3.06M

✓ \$0.50M above the \$2.56M target

Target = +20% on 2025 merch profit

Levers

LEVER	FULL POTENTIAL	CONFIDENCE	RISK-ADJUSTED	TIME TO IMPACT	OWNER
Discount redesign	\$3.71M	50%	\$1.86M	3–6 months	Pricing
Restart acquisition	\$1.63M	50%	\$0.81M	12–36 months	Marketing / Growth
Payment recovery	\$0.31M	80%	\$0.25M	1–3 months	Product / Payments
Return reduction	\$0.24M	60%	\$0.14M	6–12 months	Operations

Each lever is computed from the order lines using the parameters on the left. Acquisition value is at year 3, with 67% retention.

- Global filters don't apply here
- Every number updates live from the parameters

VALIDATE (ALL FILTERS CLEARED)

At defaults: $\$1.86\text{M} + \$0.81\text{M} + \$0.25\text{M} + \$0.14\text{M} = \$3.06\text{M}$ vs. $\$2.56\text{M}$ target

Scenario formulas and data caveats

Scenario Planner fields

FIELD	FORMULA
Scenario Discount	<code>IF [discount_pct] < [p_Floor] THEN 0 ELSEIF [discount_pct] > [p_Cap] THEN [p_Cap] ELSE [discount_pct] END</code>
Scenario Qty	<code>IF [discount_pct] >= 0.27 AND [Scenario Discount] < 0.23 THEN [quantity]*1.88/3.31 ELSEIF [discount_pct] < 0.23 AND [Scenario Discount] >= 0.27 THEN [quantity]*3.31/1.88 ELSE [quantity] END</code>
Scenario Profit	<code>[Scenario Qty]*[unit_price]*(1-[Scenario Discount]) - [Scenario Qty]*[unit_cost]</code>
Pricing lever / yr	<code>(SUM([Scenario Profit]) - SUM([merch_profit])) / 5</code>
Payment lever / yr	<code>SUM(IF [is_payment_failure] THEN [order_merch_profit] END) / 5 * [p_Recovery]</code>
Returns lever / yr	<code>SUM(IF [is_returned] THEN [order_merch_profit] END) / 5 * [p_Return Cut]</code>
Acquisition lever (yr 3)	<code>[p_New Customers] * (1 + 0.67 + 0.67^2) * [Profit per Active Customer]</code>
Retention	<code>SUM(INT([active_prior_year])) / LOOKUP(COUNTD([customer_id]), -1), along year</code>

Caveats to footnote on the dashboard

DATA ISSUE	WHERE	HANDLING
Returned lines show a 0% discount (refund likely wipes it)	Pricing	Filter <code>in_discount_analysis = True</code> ; footnote
2021 is the first year, so every customer looks "new"	Overview, Growth	Start new-customer charts at 2022
Dataset's own <code>profit</code> column doesn't reconcile	Everywhere	Use <code>merch_profit</code> only
Flat active base despite ~0 acquisition and 67% retention	Growth	Footnote: implies reactivation; confirm definition
Identical behavior across categories and segments	Pricing, Channels	About page: data appears synthetic; directional

Build order: data source and relationships → core ratio fields and parameters → validate totals → sheets → dashboards → actions → QA (check each filter on each page, colorblind preview) → publish to Tableau Public → add the link and a screenshot to the README.